

SafeWall: Risk-Aware Job Walltime Recommendation for HPC Scheduling

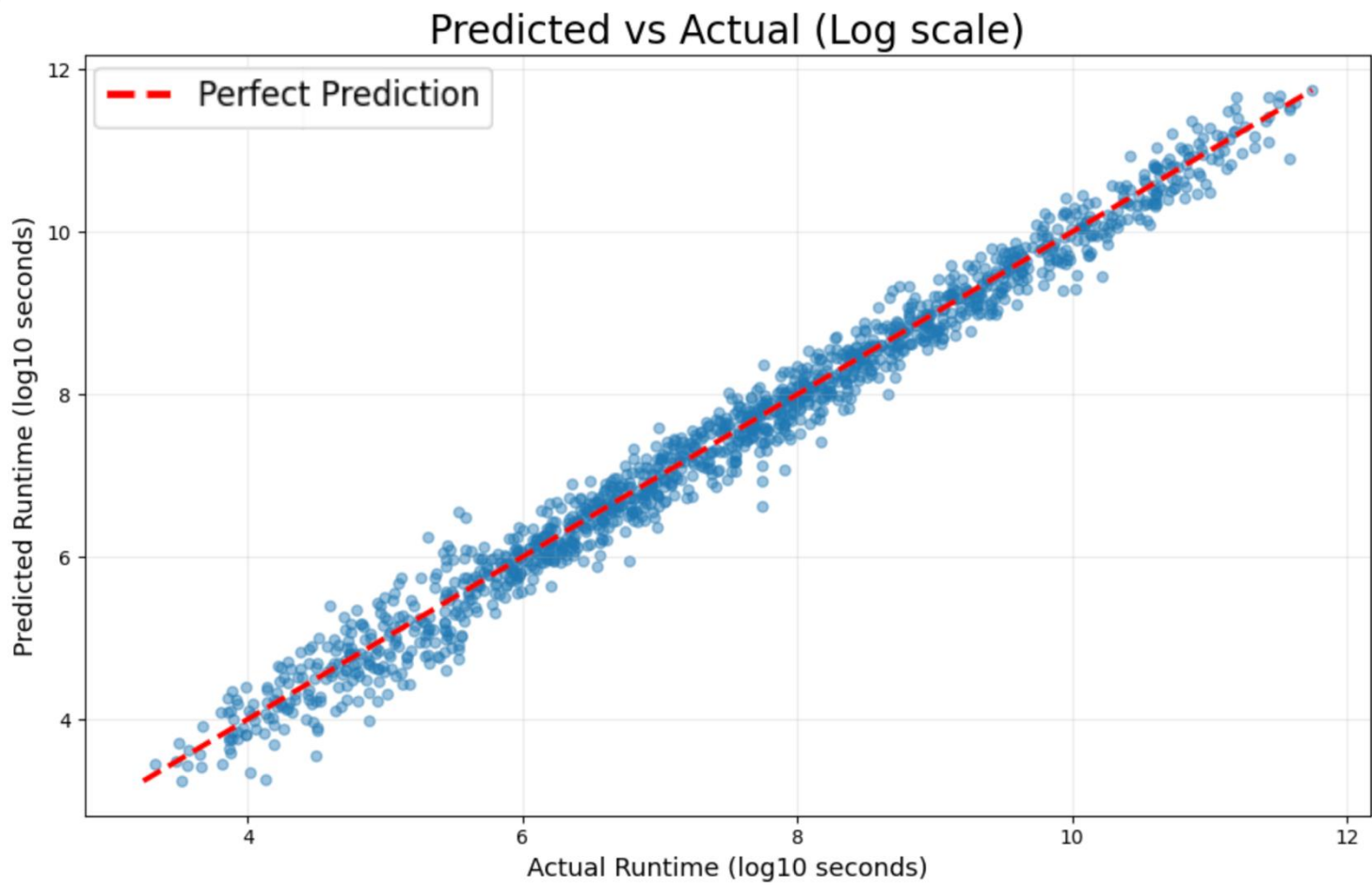
Kanglin Xu, Michael E. Papka, Zhiling Lan

Motivation

HPC schedulers treat walltime as a harddeadline. User often request conservative walltimes to avoid job termination, but overly large requests reduce backfilling efficiency and overall utilization.

Machine Learning Model Prediction

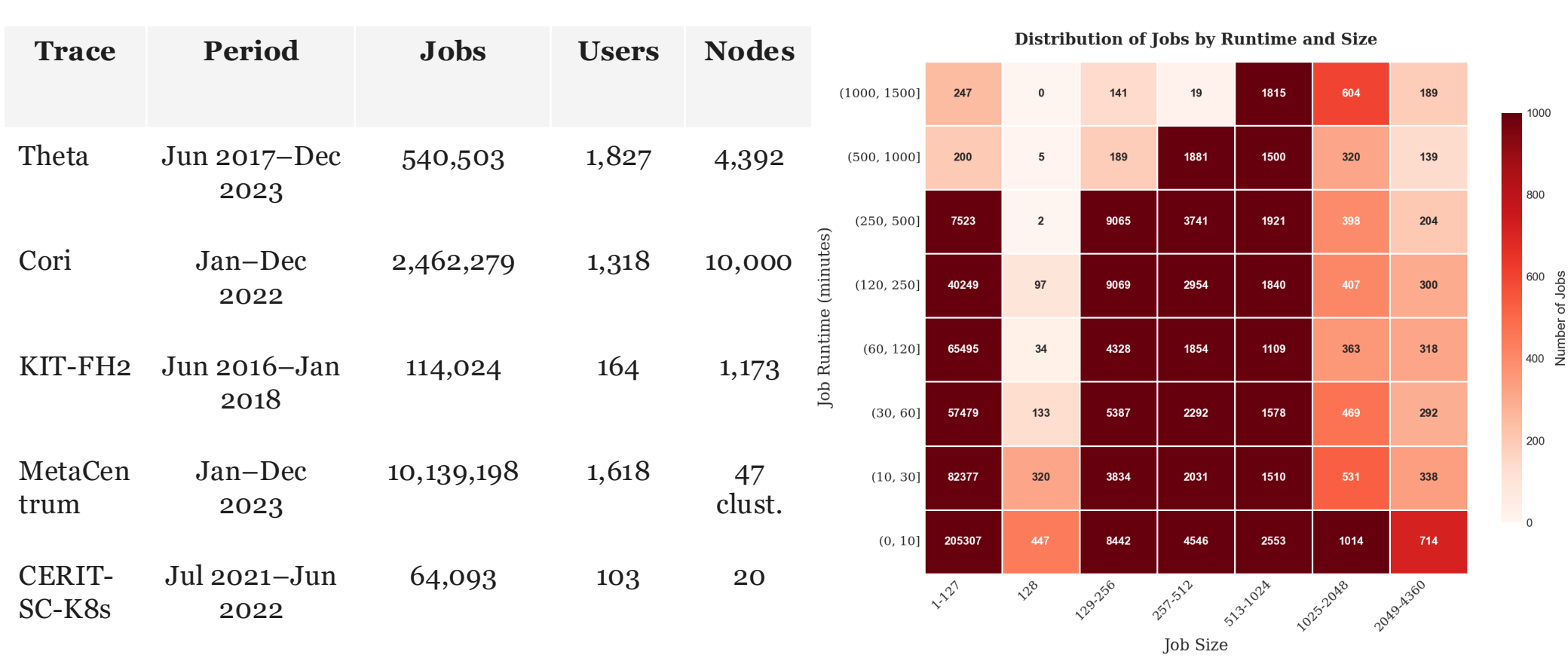
For Machine Learning predictive model that achieves high accuracy, it suffers from underestimation about half of jobs. In HPC, these underestimates may cause premature job termination.



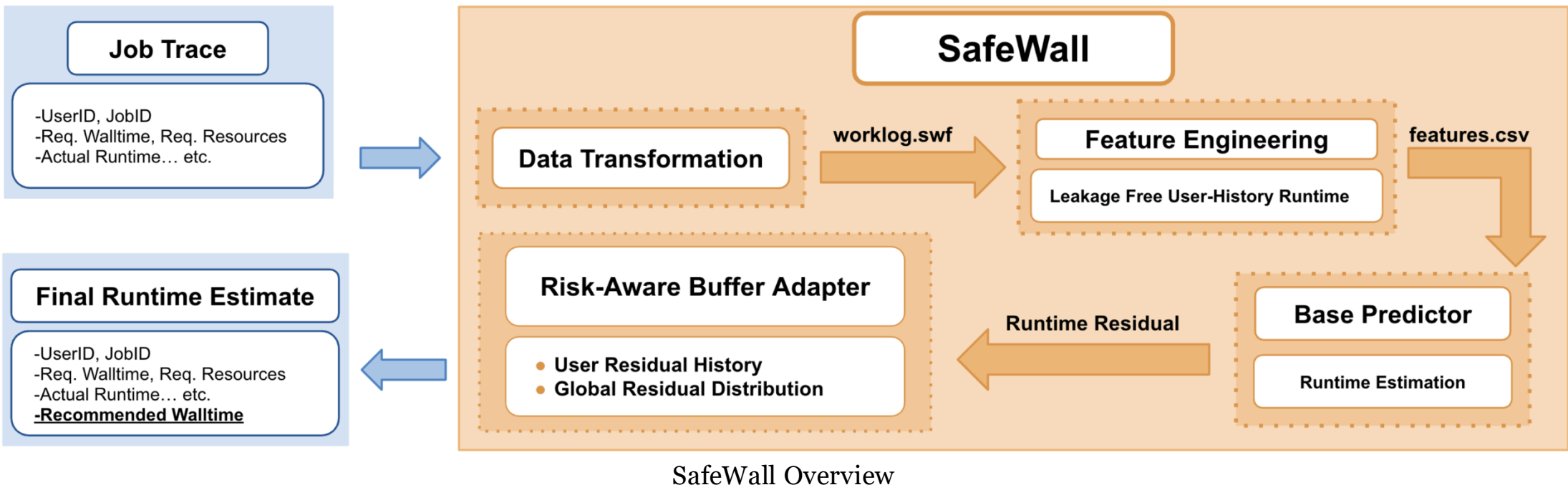
Data

We use the Theta trace (**540K jobs, 1,827 users, 4,392 nodes**) plus four public traces: **Cori, KIT-FH2, MetaCentrum, and CERIT-SC-K8s**.

Worklog Data Characteristics



Methodology



Base Predictor

For the base predictor, we train learning-based models to estimate each job’s runtime from scheduler-visible and user-history features.

We use **XGBoost** as the main point predictor and also evaluate **Quantile Gradient Boosting** as a conservative baseline. The base predictor produces the initial runtime estimate $\hat{y}_i$, which is then passed to SafeWall’s risk-aware buffer adapter.

Risk Aware Buffer

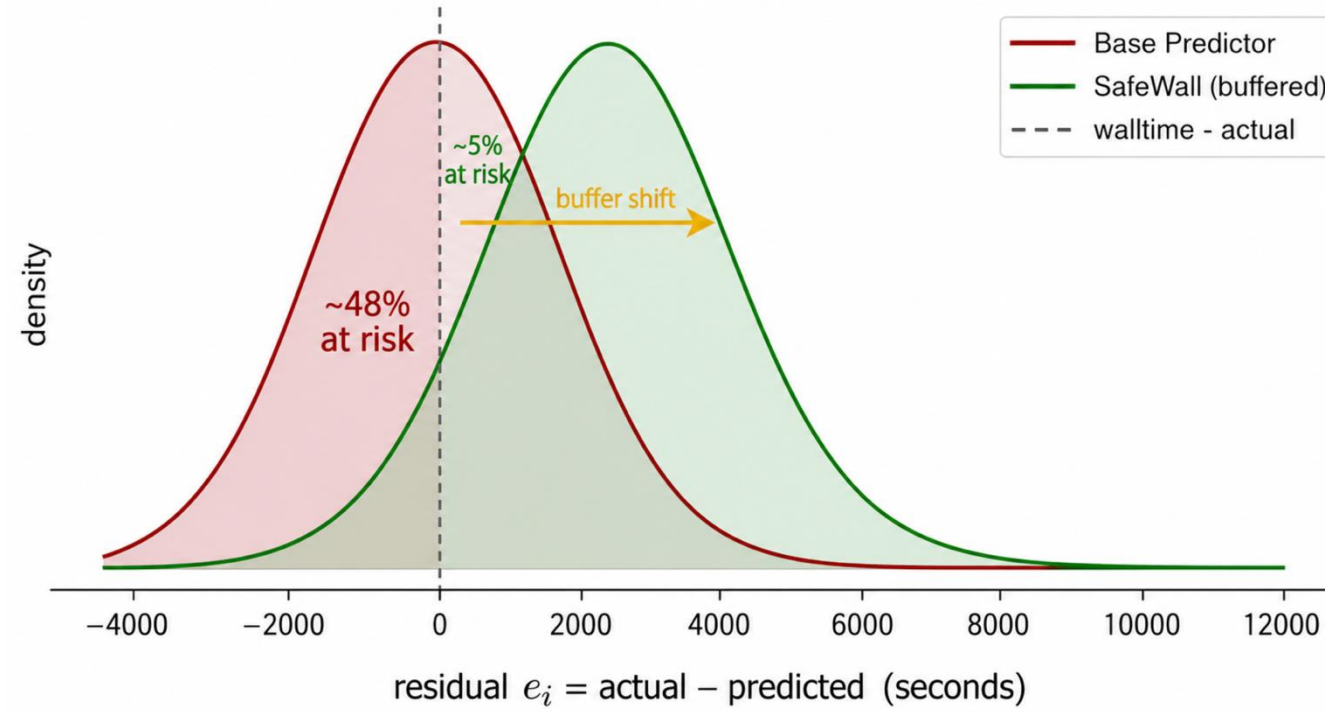
User-Based Buffer

SafeWall adjusts the buffer using each user’s past prediction errors. Users with frequent underestimation receive larger buffers, while users with limited history are pulled toward a global baseline.

User residual history + global residual distribution → personalized safety buffer.
Final recommendation: $\tilde{y}_i = \hat{y}_i + b_i$.

Data Preprocessing

Feature Group	What We Compute
Lag runtime features	Previous runtime, second previous runtime, and mean of the last two runtimes
Request accuracy	$a_j = \min(w_i a_{ij} / y_j, y_j / wall_i)$, measuring how close each prior walltime request was to actual runtime
User summaries	Mean and max request accuracy
Runtime summaries	Mean runtime, max runtime, recent mean/max, and 25th percentile of prior successful runtime
Recent-history features	Summary features over the last 10 completed jobs when available



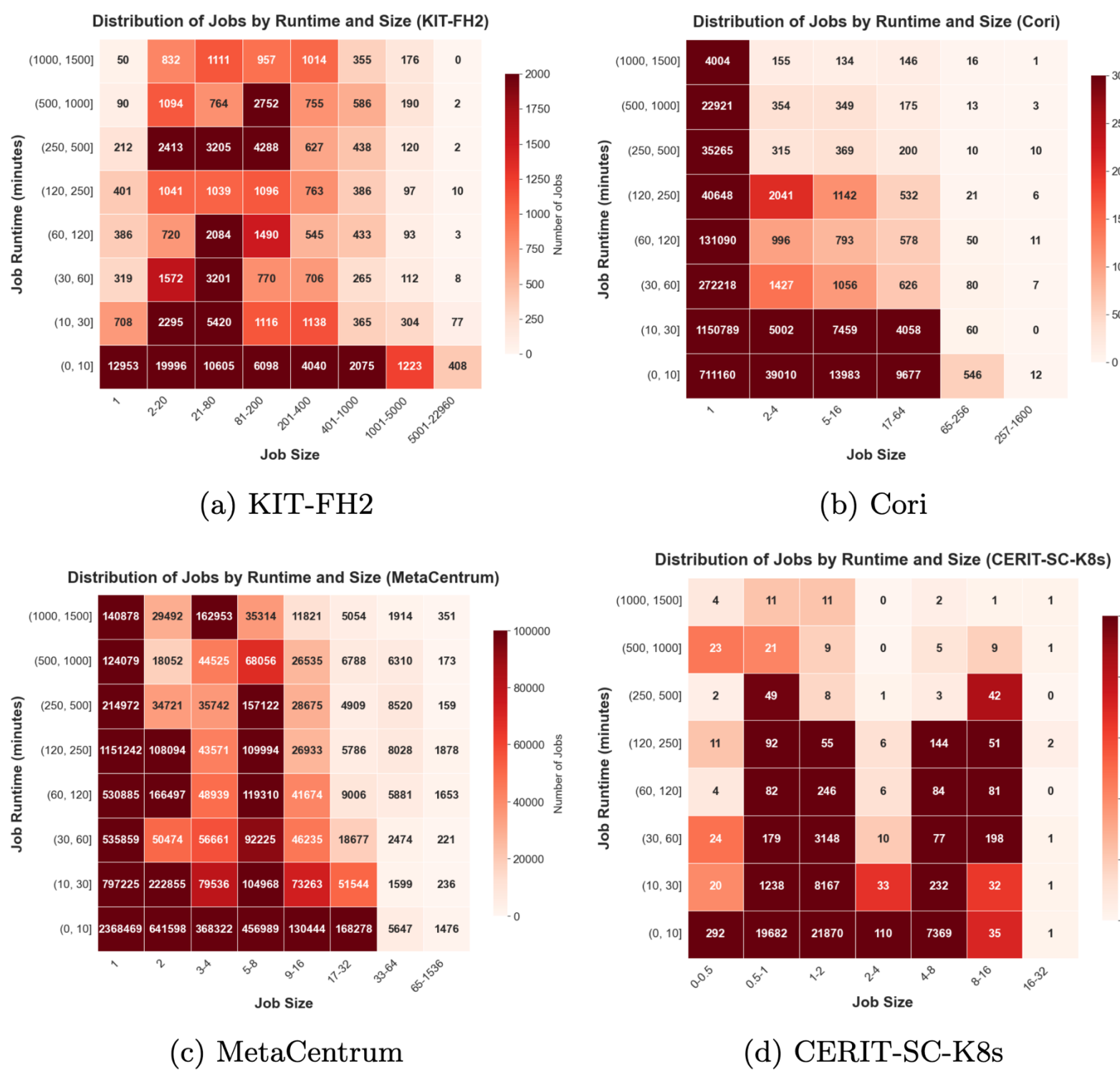
Result

Performance on Theta

Method	UR (%)	Accuracy (%)	RMSE (s)	MAE (s)	R ²
XGBoost	47.85	97.20	961.44	57.34	0.9835
XGBoost + SafeWall	4.31	90.27	963.89	91.81	0.9834
Quantile GBR	5.56	85.44	685.90	159.20	0.9916
Quantile GBR + SafeWall	3.46	87.70	770.25	147.55	0.9895
Percentage buffer	1.06	82.83	1808.66	547.43	0.9418
Fixed-value buffer	7.05	89.91	961.85	77.54	0.9835
Piecewise buffer	2.59	82.49	1484.90	384.39	0.9608

Cross System Validation

Trace	#Jobs	XGBoost UR (%)	XGBoost Acc (%)	SafeWall UR (%)	SafeWall Acc (%)
Cori	2,465,286	47.92	93.47	5.06	83.21
KIT	114,355	49.23	90.13	6.79	74.76
CERIT	64,268	49.11	92.14	5.00	83.02
MetaCentrum	10,145,231	48.16	84.03	4.92	68.64



• SafeWall consistently cuts **UR** by **about 10×** across production and public traces.

• Theta results show **large risk reduction** with nearly preserved predictive fit.

• Workload heterogeneity across Cori, KIT, CERIT, and MetaCentrum supports the need for **risk-aware recommendation**.

• The method remains effective across **very different runtime–size patterns**.

• Overall, SafeWall makes walltime recommendation **safer and more deployable**.

Discussion

- SafeWall is **model-agnostic** and can wrap different base predictors without modifying the predictor itself.
- The method turns walltime recommendation into an explicit **risk-control** problem, not just an accuracy problem.
- Cross-trace results suggest strong robustness under **workload heterogeneity**.
- Personalized buffering helps address user-specific underestimation behavior.

Limitations

- SafeWall depends on historical residuals, so performance may weaken under workload drift.
- Benefits are smaller for **cold-start** users or users with limited job history.
- Lower underestimation risk may come with slightly less aggressive recommendations.
- Results are based on trace evaluation rather than a fully deployed live scheduler.

Future Work

- **Adaptive recalibration** under changing workloads
- **Conditional buffers** that depend on job or user characteristics
- **Scheduler-in-the-loop evaluation** in realistic production settings